

AI Voice Support Bot

Multilingual Customer Support for Airtel — Powered by Sarvam AI

D2C Customer Support · Voice Bot + Agentic Workflow · Hindi, English & Marathi

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github.com/nrvm94/sarvam-telecom-bot

The Problem

India's telecom support infrastructure is failing customers at scale

950K

inbound support calls
per day — Airtel, all channels

₹28–35

cost per human agent
interaction

8–12 min

average customer wait
during peak hours

71%

abandon DTMF IVR
before resolution (TRAI 2024)



Language Barrier

68% of subscribers prefer regional languages. Legacy IVR systems offer Hindi/English only — failing Tier-2 and Tier-3 customers entirely.



24/7 Gap

Call centres staffed ~18 hours/day. Customers with billing disputes or outages at 2 AM have no recourse.



Repetition Fatigue

Average customer repeats their query 2.3× due to misrouting and poor NLU in legacy IVR systems.



Cost Trajectory

With 485M subscribers, even 1% call volume growth adds ₹130 crore/year in agent costs — unsustainable at current rates.

Why Voice AI?

Voice is the natural interface for India's next 300 million connected users

500M+

Indians use voice search on mobile
(Google India 2024)

62%

of Tier-2/3 first-time internet users
prefer voice over typing

78%

of Airtel Tier-1 queries are fully
resolvable without a human agent

AI Voice Bot vs. Status Quo		
Metric	Today	With AI Voice Bot
Wait time	8–12 min on hold	< 3 seconds
Availability	6 AM – 12 AM only	24 / 7 / 365
Languages	Hindi + English only	Hindi, English, Marathi
Cost per call	₹28–35	₹0.40
Consistency	Variable (human error)	100% script adherence
Scale	Staffing-constrained	Unlimited, zero incremental cost

Why Sarvam AI?

The only AI stack purpose-built for India's languages, regulation, and infrastructure

Capability	Google / AWS	Yellow.ai	OpenAI GPT-4o	Gnani.ai	Sarvam AI ✨
Hindi-English code-mixing	Partial	Limited	Limited	Partial	✓ Native (trained on Hinglish)
Indian accent STT	Partial	3rd-party	Not fine-tuned	✓ Yes	✓ 500M+ Indian voices
STT latency (India)	800–1200ms	800–1000ms	700–1200ms	500–800ms	≤ 300 ms
Own India-trained LLM	✗ Generic	✗ GPT wrapper	✗ Own but US	✗ Wrapper	✓ sarvam-105b India-first
Data sovereignty / TRAI	US servers	US servers	US servers	India	✓ In-country processing
On-premise deployment	✗	✗	✗	Partial	✓ Private cloud / on-prem
Indian language breadth	8 langs	12 langs	8 langs	6 langs	22+ Indian languages
Model lock-in risk	High	High	Very High	Medium	✓ OpenAI-compatible API

For Airtel: TRAI compliance + on-premise deployment + Hinglish code-mixing is a non-negotiable stack — only Sarvam delivers all three natively.

Sarvam AI — Enterprise Proof Points

Deployed at scale across BFSI, Government, and Service industries

BFSI • Insurance

SBI Life Insurance

**Millions of
policy calls automated**

AI voice agents handle policy inquiries, renewal reminders, and claims status in 10+ Indian languages — 24/7, at a fraction of human agent cost.

BFSI • Consumer Lending

Tata Capital

**3× increase in
customer engagement**

Multilingual voice agents across consumer loan products — personalised, segment-specific conversations that break language barriers at scale.

BFSI • Rural Finance

Mahindra Finance

**Rural-first
voice collections**

Multilingual EMI reminder agent reaching farmers and rural borrowers in their native language — populations unreachable via text or app.

D2C • On-demand

Urban Company

**Post-service
voice follow-up**

Automated voice-first customer feedback and support resolution — reducing manual callback queues for service quality escalations.

Government

Skill India / NABARD

**50,000+ farmer
feedback calls**

Conversational AI collecting structured citizen feedback for Farmer Field School programmes in Maharashtra — government-scale regional language reach.

Why this matters for Airtel

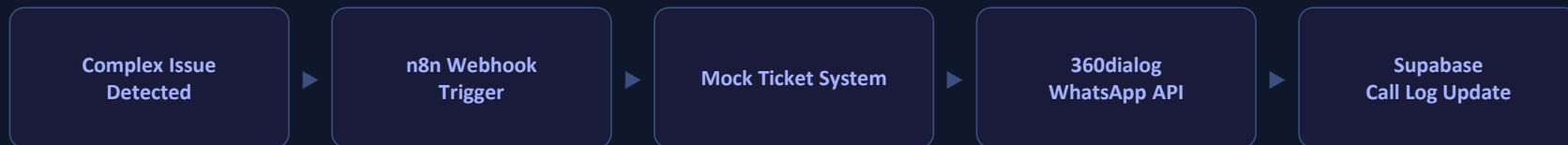
Every deployment above mirrors Airtel's challenge: high-volume, multilingual, cost-sensitive. Sarvam's stack is already proven at this scale.

What We Built

Full-stack solution: Voice Bot + Agentic Escalation Workflow



ESCALATION PATH (Agentic Workflow — triggers when bot cannot resolve)



Stack:

React 18 + Vite

FastAPI + Python

ChromaDB (RAG)

Supabase (PostgreSQL)

n8n Automation

360dialog WhatsApp

GitHub: github.com/nrvn94/sarvam-telecom-bot · Full source code, .env.example, setup guide, architecture docs

ROI & Business Case

Realistic enterprise cost model with scenario-based adoption assumptions

Stated assumptions: 950K calls/day, 78% Tier-1 automatable, ₹28/call human cost, ₹0.40/call AI cost, 6-month linear ramp to stated adoption level

₹0.40

cost per AI-handled call
vs ₹28–35 human agent

70×

reduction in cost
per interaction

< 5 weeks

break-even at conservative
20% adoption rate

Investment (Year 1)

Telephony / SIP integration	₹2–3 Cr
LLM fine-tuning + KB ingestion	₹1.5–2 Cr
CRM + billing integration	₹2–3 Cr
Cloud infra + GPU + redundancy	₹2–3 Cr
Security + compliance + change mgmt	₹2–3 Cr

Total setup cost ₹10–14 Cr

Year 2+ OpEx: Sarvam API + infra + support = ₹11–16 Cr/year

Returns by Adoption Scenario

	Conservative (20%)	Base Case (50%)	Optimistic (80%)
Calls automated / day	148,000	370,000	592,000
Annual saving (₹27.60/call)	₹149 Cr	₹373 Cr	₹596 Cr
Net Year 1 (after all costs)	₹125–138 Cr	₹349–362 Cr	₹572–585 Cr
Break-even timeline	~5 weeks	~2 weeks	~1 week

Additional upside: 1% churn reduction = ₹85 Cr ARR retained · Upsell via bot: +₹2–5 ARPU · Call transcripts = continuous product intelligence at zero extra cost

Limitations & 90-Day Enterprise Rollout Plan

Honest assessment of current state — and a structured path to full-scale deployment

System Integration

Requires CRM, billing & IVR integration. Responses are generic without live subscriber data.

Model Fine Tuning

To provide on-brand, contextually accurate responses at scale

Compliance & Auth

TRAI data residency and OTP caller authentication required before exposing account data.

KB Richness

Bot currently works on limited data set. Accuracy mirrors richness of data.

Change Management

Agent and stakeholder buy-in is a deployment risk as significant as any technical dependency.

90-DAY ENTERPRISE ROLLOUT PLAN

Days 1–30 · Scope & Commit

Activities:

Joint discovery · baseline measurement · success metrics workshop · stakeholder alignment (IT, CISO, CFO) · MSA signed · KB audit

Customer commits:

Executive sponsor · IT point of contact · KB documentation · Success metrics signed off

Deliverable:

Signed scope + agreed KPIs · Architecture blueprint · KB v1 validated

KPIs:

Signed scope document · Architecture blueprint approved · KB v1 ingested and validated

Days 31–60 · Pilot & Validate

Activities:

Telephony + CRM integration · security & TRAI audit · soft launch 2–5% volume · weekly review cadence · failure sprint cycles

Customer commits:

2 IT resources · contact centre supervisor · weekly review attendance

Deliverable:

Live pilot running · Week-6 CXO deck with real CSAT + cost-per-call data

KPIs:

Containment rate · First call resolution · Escalation rate · Latency

Days 61–90 · Scale & Prove

Activities:

Expand to 15–25% across 2 circles · query type + language expansion · commercial review · board presentation · Phase 2 SOW

Customer commits:

CXO for board presentation · Contact centre team feedback session

Deliverable:

Board-ready ROI report on live data · Phase 2 business case · Full-scale approval

KPIs:

Cost per call vs baseline · NPS delta · Automation rate · Monthly ₹ saved